

TrES-3b

R-1785

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210525 · created 2026-08-03T00:07:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459359.7125 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.039
Transit depth (Rp/Rs)^2	1.96 +/- 1.09 [%]
Orbital Inclination (inc)	82.62 +/- 2.08
Airmass coefficient 1 (a1)	1.266 +/- 0.024
Airmass coefficient 2 (a2)	-0.1525 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	None
Optimal Aperture	3.37
Optimal Annulus	12.39
Transit Duration (day)	0.054 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.039 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.054 d vs 0.0567 d (deviation 0.11σ)
Mid-time O–C	-4.4 min
Depth SNR	3.1
Residual scatter	1.89 %

Light curve

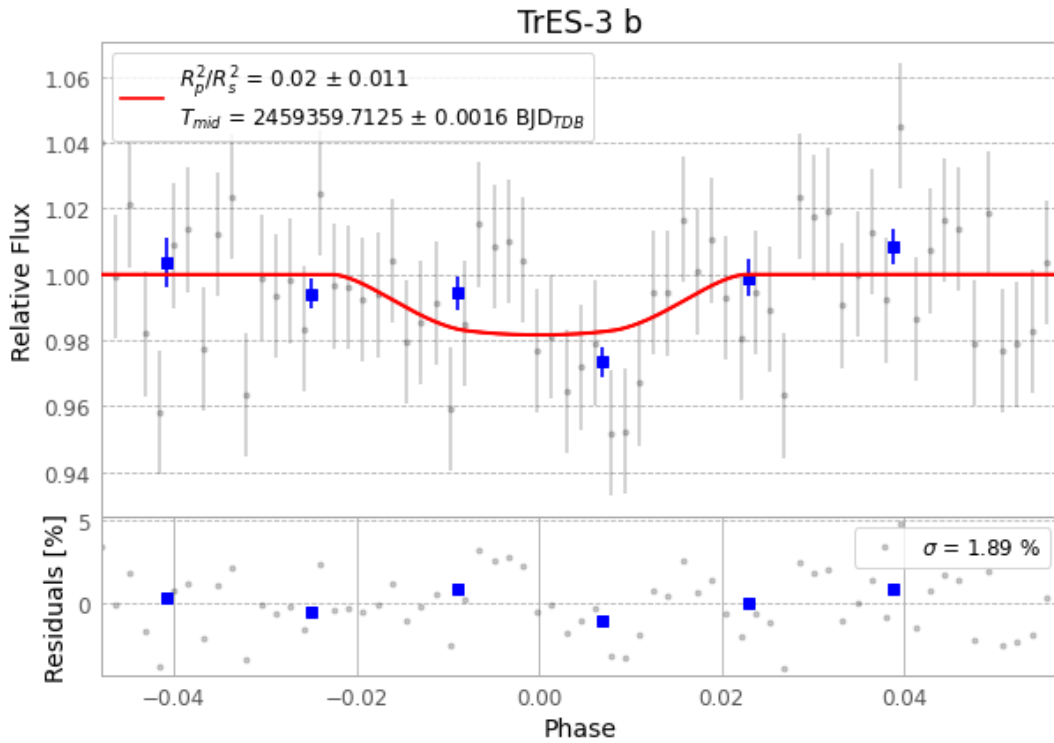


Figure 1 — FinalLightCurve_TrES-3 b_2021-05-24.png (SHA-256 cdbb5d9af832c650...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c31a61118d78667...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3210525033012.FITS → TRES-3210525064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210525.log (b38c31c5520d...), FinalLightCurve_TrES-3 b_2021-05-24.png (cdbb5d9af832...), FinalLightCurve_TrES-3 b_2021-05-24.csv (45fdb1da04a1...)

Compute resources

Wall time 340.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:07Z.